Lesson 24 Inclusion-Exclusion Principle

Formula for the cardinality of a union of sets

Simplest case U: universal set ; $A, B \subseteq U$

Union Rule $|A \cup B| = \underbrace{|A| + |B|}_{\text{inclusion}} - \underbrace{|A \cap B|}_{\text{exclusion}}$

double counted

2 Corollaries

① Addition Rule : When A and B are disjoint, then $A \cap B = \phi$ and $|A \cap B| = 0$

$$\boxed{|A \cup B| = |A| + |B|}$$

complement of A

② Subtraction Rule : A and $\overline{A}$ are disjoint

By the add^n rule, $|\underbrace{A \cup \overline{A}}_{U\text{ : unversal}}| = |A| + |\overline{A}|$

$$\rightarrow \boxed{|\overline{A}| = |U| - |A|}$$

Ex $U = \{1, 2, \ldots, 100\}$

A = multiples of $3 = \{3, 6, 9, \ldots, 99\}$

B = " " $5 = \{5, 10, 15, \ldots, 100\}$

$A \cup B$: multiples of 3 or 5

$|A| = \lfloor \frac{100}{3} \rfloor$ (floor) $= 33$

$|B| = \lfloor \frac{100}{5} \rfloor = 20$

$|\underbrace{A \cap B}_{\text{multiples of 15}}| = \lfloor \frac{100}{15} \rfloor = 6$

multiple of 3 and 5 (p, q)

$\updownarrow$

multiple of 15 (pq)

$|A \cup B| = |A| + |B| - |A \cap B| = 33 + 20 - 6 = 47$

$|\overline{A \cup B}| = |\overline{A} \cap \overline{B}|$ (neither multiple of 3 nor 5)

De Morgan $\quad x = |U| - |A \cup B|$

$$= 100 - 47 = 53$$

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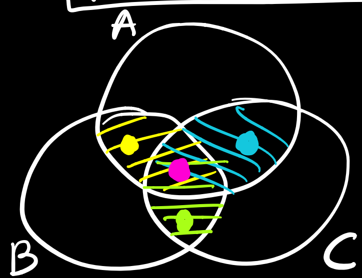
3 sets : $A, B, C \subseteq U$ (universal)

$$\boxed{|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|}$$



Example 2 in notes

General case : $A_1, A_2, \dots, A_n \subseteq U$

$$\boxed{|A_1 \cup A_2 \cup \cdots \cup A_n| = \sum_i |A_i| - \sum_{i,j} |A_i \cap A_j| + \sum_{i,j,k} |A_i \cap A_j \cap A_k| - \cdots + (-1)^{n+1} |A_1 \cap A_2 \cap \cdots \cap A_n|}$$

Version 2 (Complement) (General DeMorgan- Induction)

$$|\overline{A_1 \cup A_2 \cdots \cup A_n}| = |\bar{A}_1 \cap \cdots \cap \bar{A}_n|$$

$$\underbrace{|\bar{A}_1 \cap \bar{A}_2 \cap \cdots \cap \bar{A}_n|}_{\text{LHS}} = |U| - |A_1 \cup A_2 \cup \cdots \cup A_n|$$

$$\text{LHS} = \underbrace{|U| - \sum_i |A_i| + \sum_{i,j} |A_i \cap A_j| - \sum_{i,j,k} |A_i \cap A_j \cap A_k| + \cdots + (-1)^n |A_1 \cap A_2 \cap \cdots \cap A_n|}_{\text{RHS}}$$

Proof LHS counts the # of elements which are **not** in any A_i. We will show that RHS counts the same elements.

- An element x which is not in any A_i contributes "1" to RHS:
  - x's contribution to $|U|$ is 1
  - x's contribution to the other terms on RHS is 0

  Net contribution of x to RHS is 1.
- An element y which is in at least one A_i contributes "0" to RHS: Suppose y is in exactly m A_i's
  - y's contribution to $|U|$ is 1
  - y's " to $\sum_i |A_i|$ is m
  - y's . . . . to $\sum |A_i \cap A_j|$ is $\binom{m}{2}$

    # of 2-set intersections with y in each set.
  - y's . . . to $\sum |A_i \cap A_j \cap A_k|$ is $\binom{m}{3}$

    ⋮
  - y's . . . (intersections of m sets) is $\binom{m}{m}$

y's net contribution to RHS:

$$1 - m + \binom{m}{2} - \binom{m}{3} + \cdots \binom{m}{m} = 0$$

$\binom{m}{0}$ $\binom{m}{1}$

proved earlier

Exercise in Combinatorial proof:

the # of even-numbered subsets

= . . . . odd-numbered . . .

$$\binom{m}{0} + \binom{m}{2} + \binom{m}{4} + \cdots = \binom{m}{1} + \binom{m}{3} + \cdots$$